

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

Overview

CTS Corporation ("CTS", "we", "our" or "us") is a leading designer and manufacturer of products that Sense, Connect and Move. Our vision is to be a leading provider of sensing and motion devices as well as connectivity components, enabling an intelligent and seamless world. These devices are categorized by their ability to Sense, Connect or Move. Sense products provide vital inputs to electronic systems. Connect products allow systems to function in synchronization with other systems. Move products ensure required movements are effectively and accurately executed. We are committed to achieving our vision by continuing to invest in the development of products and technologies within these categories.

We manufacture sensors, actuators, and electronic components in North America, Europe, and Asia. CTS provides engineered products to OEMs and tier one suppliers in the aerospace and defense, industrial, information technology, medical, telecommunications, and transportation markets.

There is an increasing proliferation of sensing and motion applications within various markets we serve. In addition, the increasing connectivity of various devices to the internet results in greater demand for communication bandwidth and data storage, increasing the need for our connectivity products. Our success is dependent on the ability to execute our strategy to support these trends. We are subject to challenges including periodic market softness, competition from other suppliers, changes in technology, and the ability to add new customers, launch new products or penetrate new markets.

Results of Operations: Year Ended December 31, 2019, versus Year Ended December 31, 2018

(Amounts in thousands, except percentages and per share amounts):

The following table highlights changes in significant components of the Consolidated Statements of Earnings for the years ended December 31, 2019, and December 31, 2018:

	Years Ended December 31,		Percent Change	Percent of Net Sales	
	2019	2018		2019	2018
Net sales	\$ 468,999	\$ 470,483	(0.3)	100.0	100.0
Cost of goods sold	311,424	305,510	1.9	66.4	64.9
Gross margin	157,575	164,973	(4.5)	33.6	35.1
Selling, general and administrative expenses	70,408	73,569	(4.3)	15.0	15.6
Research and development expenses	25,967	25,304	2.6	5.5	5.4
Restructuring charges	7,448	5,062	47.1	1.6	1.1
(Gain) on sale of assets	(63)	—	—	—	—
Total operating expenses	103,760	103,935	(0.2)	22.1	22.1
Operating earnings	53,815	61,038	(11.8)	11.5	13.0
Other (expense) income, net	(3,549)	(2,935)	20.9	(0.8)	(0.6)
Earnings before income tax	50,266	58,103	(13.5)	10.7	12.4
Income tax expense	14,120	11,571	22.0	3.0	2.5
Net earnings	\$ 36,146	\$ 46,532	(22.3)	7.7	9.9
Diluted earnings per share:					
Diluted net earnings per share	\$ 1.09	\$ 1.39			

Sales were \$468,999 for the year ended December 31, 2019, a decrease of \$1,484, or 0.3% from 2018. Sales to transportation markets decreased \$1,119 or 0.4%. Sales to other markets decreased \$364 or 0.2%. The QTI acquisition, which was completed in July 2019, added \$9,252 in sales for the year (see Note 19). Changes in foreign exchange rates decreased sales by \$5,135 year-over-year due to the U.S. Dollar appreciating compared to the Chinese Renminbi and Euro.

Gross margin as a percent of sales was 33.6% in 2019 versus 35.1% in 2018. The decrease in gross margin was driven primarily by lower volume, foundry inefficiency in our ceramics manufacturing operations, labor and commodity cost increases, a one-time purchase accounting step-up in the value of finished goods inventory acquired with our QTI acquisition, and an unfavorable impact from currency movements. These were partially offset by cost improvement projects as well as savings from our manufacturing transition project.

Selling, general and administrative ("SG&A") expenses were \$70,408, or 15.0% of sales for the year ended December 31, 2019, versus \$73,569 or 15.6% of sales in the comparable period of 2018. The 2019 SG&A costs include amortization of intangibles and other operating costs associated with the QTI acquisition as well as higher environmental expenses, which were offset by

lower short-term incentive compensation, the impact of cost reduction actions taken in the fourth quarter, and other cost controls during the year.

Research and development expenses were \$25,967 or 5.5% of sales in 2019 compared to \$25,304 or 5.4% of sales in 2018.

Restructuring charges were \$7,448 for year ended December 31, 2019. The charges were mainly for building and equipment relocation, severance, and asset impairment charges related to the restructuring of certain operations, as well as a lease termination fee related to a property lease we acquired in the QTI acquisition. Restructuring charges were \$5,062 in 2018.

Operating earnings were \$53,815, or 11.5% of sales in 2019, compared to \$61,038, or 13.0% of sales in 2018 as a result of the items discussed above.

Other income and expense items are summarized in the following table:

	Years Ended December 31,	
	2019	2018
Interest expense	\$ (2,648)	\$ (2,085)
Interest income	1,737	1,826
Other (expense) income	(2,638)	(2,676)
Total other (expense) income, net	\$ (3,549)	\$ (2,935)

Interest expense increased mainly as a result of an increase in debt related to the QTI acquisition. Other expense in 2019 was principally driven by foreign currency translation losses, mainly due to the appreciation of the U.S. Dollar compared to the Chinese Renminbi and Euro, as well as an increase in pension expense.

	Years Ended December 31,	
	2019	2018
Effective tax rate	28.1%	19.9%

The effective income tax rate in 2019 was 28.1% compared to 19.9% in the prior year. The tax rate in 2019 was higher than the U.S. statutory federal tax rate primarily due to foreign earnings that are taxed at higher rates, the impact of taxes on unremitted earnings and unfavorable increases to reserves. The tax rate in 2018 was favorably impacted by a one-time rate change benefit related to the Tax Cuts and Jobs Act of 2017 resulting from the election of tax accounting method changes to accelerate deductions on the 2017 tax return, partially offset by a one-time withholding tax on repatriation of earnings from one of our foreign subsidiaries that was completed during the year to enable the use of tax credits due to expire in 2018.

Net earnings were \$36,146 or \$1.09 per diluted share for the year ended December 31, 2019, compared to earnings of \$46,532 or \$1.39 per diluted share in the comparable period of 2018.